NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 20	**QUIZ-3**	**Time: 20 Minutes**

Subject: Probability and Statistics **Section: C**
Instructor: Hazber Samson Date: 17-03-2026
Student Name_____ Roll No_____

Q1. **(4+4+4+4+4=20)**

Let X denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

		x		
$f(x,y)$		1	2	3
y	1	0.05	0.05	0.10
	3	0.05	0.10	0.35
	5	0.00	0.20	0.10

(a) Evaluate the marginal distributions of X and Y.
(b) Find $P(Y = 3 \mid X = 2)$ and $P(X + Y \leq 3)$
(c) Check whether X and Y are independent or not.
(d) Find $E(XY)$, $E(X)$, $E(Y)$
(e) Find $COV(X,Y)$ *and* $Corr(X,Y)$

GOOD LUCK !